Ghosts, Geometry, and Reality in Fourth-Order Quantum Theories

A guided introduction to positive metrics, Krein completions, residual cohomology, and interaction obstructions

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Abstract

The phrase "higher-derivative ghost problem" combines three different questions: which real-form completions exist for the local free dynamics, which classes survive local and residual gauge reduction, and whether a chosen completion persists under interaction. This introduction organizes an eight-paper programme around those three levels. Papers I–III classify the Pais–Uhlenbeck and scalar-field completions, distinguish positive metrics from Krein real forms and vacuum covariance, and explain the Jordan limit. Paper IV adds local gauge reduction and Lorentz covariance in Einstein–Weyl gravity. Papers V–VI form the interaction branch: resonant branch conversion obstructs the canonical analytic positive metric in the scalar field and in the open gravitational channel, without excluding every indefinite, nonlocal, or nonperturbative completion.

Papers VII–VIII form a separate pure-Weyl branch on the conformal cylinder. Both use the selected closed-universe problem in which all fifteen residual conformal transformations, including cylinder time translation, are constraints. Paper VII is the state-side theorem. It derives the residual BV–BFV complex, performs the positive-frequency polarization, and computes the absolute residual Chevalley–Eilenberg group

$$H^4_{\mathrm{res}} = \mathrm{span}\{[W^2_+], [W^2_-]\}, \qquad \mathrm{sig}\, G_{\mathrm{res}} = (2,0), \qquad G_{\mathrm{res}} = I_2.$$

These are ghost-dressed deformation or vertex classes, not one-particle gravitons. Paper VIII is the covariant-side theorem. It constructs a support-local curvature prolongation and causal chain homotopies for the complete free metric BV complex, proves the compact-to-global causal quasi-isomorphism, recovers the residual endpoints, transports the $SO(4,2)$ action, and identifies the Green, Cauchy, energy-mode, and residual pairings. It imports Paper VII's cohomology theorem and does not recompute the CE complex; consequently

$$H^4_{\mathrm{cov}} \cong H^4_{\mathrm{res}} = \mathrm{span}\{[W^2_+], [W^2_-]\}, \qquad \mathrm{sig}\, G_{\mathrm{cov}} = (2,0), \qquad G_{\mathrm{cov}} = I_2.$$

The combined result is free and formal near the conformal cylinder: its field-side theorem is classical, while its state-side theorem uses a selected completed positive-frequency representation. It is not a positive graviton Hilbert-space theorem, a Hadamard construction, a

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nonlinear stability theorem, or a renormalized quantum-master-equation result. The programme therefore replaces a single ghost verdict by a typed architecture: local completion, residual reduction, covariant causal transport, and interaction stability are related, but none substitutes for the others.

Contents

1 Why higher derivatives create ghosts

Most familiar equations of motion contain at most two time derivatives. Newton's equation has the form

$$m\ddot{q} = F(q), \tag{1}$$

and the Klein–Gordon field equation has the form

$$(\Box + m^2)\phi = 0, \tag{2}$$

where the empty square is the standard symbol for the wave operator (d'Alembertian), $\Box = \partial^2/\partial t^2 - \nabla^2$ — the spacetime analogue of the Laplacian. Specifying a solution requires two pieces of initial data: position and velocity for a particle, or the field and its first time derivative for a field theory.

A fourth-order equation such as

$$\left(\frac{d^2}{dt^2} + \omega_1^2\right)\left(\frac{d^2}{dt^2} + \omega_2^2\right) z(t) = 0 \tag{3}$$

requires four pieces of initial data. Its solution contains two independent frequencies,

$$z(t) = A_1 e^{-i\omega_1 t} + B_1 e^{i\omega_1 t} + A_2 e^{-i\omega_2 t} + B_2 e^{i\omega_2 t}. \tag{4}$$

The extra derivatives have introduced an extra oscillator.

That sounds harmless. The difficulty is that the standard Hamiltonian formulation usually assigns opposite signs to the two oscillators:

$$H \sim H_{\omega_1} - H_{\omega_2}. \tag{5}$$

One branch then appears to carry negative energy. If one changes the quantization convention to keep the energy positive, the same branch may instead acquire negative norm. This is the traditional higher-derivative ghost problem.

The classical theorem of Ostrogradsky [1] explains why the Hamiltonian obtained from a nondegenerate higher-derivative Lagrangian is generically unbounded in its canonical momenta (see [2] for a modern review). But that theorem does not by itself answer every quantum question. The questions divide into three logically distinct levels.

Level 1 — local free dynamics and completion. Can the Hamiltonian be diagonalized? Can a positive inner product be found? Which variables count as real? What happens at the Jordan limit where distinct frequencies or masses coincide?

Level 2 — gauge and residual reduction. After the local gauge constraints have been imposed, which oscillator modes survive as genuine states or observables? On backgrounds with residual symmetries, should those symmetries be retained as global charges or treated as gauge? If they are gauged, what is the resulting BRST cohomology, and does positivity belong to propagating states, to vertex classes, or to neither?

Level 3 — interaction stability. Can a free positive metric or Krein involution be deformed when interactions are added? Can ghost parity remain conserved when particles convert into one another? Do exact energy-conserving processes obstruct the deformation? Does a doubled symmetry survive even if an individual pointed sector fails?

These levels must not be conflated. Solving the free Hamiltonian settles neither the residual gauge problem nor the interacting ghost problem. Residual reduction can remove locally propagating oscillator classes without changing their local equations, while interactions test whether a chosen free completion survives energy exchange. The research programme therefore has two branches after the free gravity classification: Papers V–VI study interaction stability, whereas Papers VII–VIII study the state-side and covariant-side residual problems at the pure-Weyl boundary.

Series map. The eight technical papers have distinct jobs.

1. Paper I reconstructs and classifies the positive free Pais–Uhlenbeck quantization.
2. Paper II characterizes its canonical metric by a minimum-distortion principle.
3. Paper III separates metric, vacuum, representation, and Jordan data for the fourth-order scalar field.
4. Paper IV adds local gauge reduction and Lorentz covariance in free Einstein–Weyl gravity and reaches the pure-Weyl boundary.
5. Paper V formulates the interaction deformation problem and identifies resonant scalar obstructions.
6. Paper VI applies that deformation complex to Einstein–Weyl gravity.
7. Paper VII computes the selected residual state cohomology at the pure-Weyl boundary.
8. Paper VIII constructs the covariant causal bridge to Paper VII's already computed residual result.

Thus Papers I–IV form the local free spine, Papers V–VI the interaction branch, and Papers VII–VIII the state/covariant residual branch. Paper VIII does not perform another Chevalley–Eilenberg calculation.

A cross-programme D-quotient validation dossier now records the scope of this last branch. Paper IX is reserved for a possible classical theorem on Taub reduction, relational clocks, and boundary dependence. The first scalar-clock gate is now a certified no-go: for one real conformally coupled scalar the invariant homogeneous mode is a branch-local oscillator clock, but its positive improved stress is incompatible with the Bach-flat vacuum cylinder; at the only compatible background, $\bar{T} = 0$, its linearized Diff×Weyl incidence vanishes. This supplies the scalar-clock scope half of the Paper IX gate. The paper is not part of the present eight-paper theorem sequence until at least one complete boundary or interaction theorem has also been certified.

> Fourth-order theories can admit several inequivalent real-form completions of one complex spectral dynamics. Local free geometry determines which completions are available; residual gauge reduction determines which state and vertex classes survive; physical interactions determine whether the surviving structures remain dynamically compatible.

Strength of the claims. The free classifications are theorems in the classes stated below. In Paper V the 3:1 and 3:2 oscillator obstruction classes are exact, the 5:1 fourth-order result is a computational proposition, and the general resonance hierarchy remains a conjecture. For the field theory what is proved is an analytic metric obstruction on a nonempty open shell subset, not a complex spectrum or the absolute nonexistence of every positive completion. In gravity, Paper VI proves first-order cubic protection and a tree-level second-order obstruction based at the canonical Paper-IV positive real form; it does not prove an all-orders no-go, complex gravitational spectrum, or failure of every nonlocal or differently pointed completion. Its Krein result concerns the natural particle-number-diagonal, cluster-multiplicative lift induced on physical BRST cohomology, not every Poincaré-covariant involution on a multiparticle space. A full-field complex spectrum and a quantum operator implementing the doubled scalar exchange remain open.

Paper VII proves its residual theorem only for the stated formal, closed-universe, positive-frequency polarization. Paper VIII proves a free classical Lorentzian causal transport theorem for the corresponding metric BV complex. Neither paper claims an integrated $SO(4,2)$ representation, a positive graviton Fock space, a BRST-compatible Hadamard state, nonlinear stability, anomaly cancellation, or interacting unitarity. Common-core commutators of unbounded generators do not by themselves provide group integrability [19].

2 The Pais–Uhlenbeck oscillator as a microscope

The Pais–Uhlenbeck (PU) oscillator [3] is the simplest model containing the full fourth-order difficulty. Its Lagrangian can be written

$$L = \frac{\gamma}{2}\left(\ddot{z}^{\,2} - (\omega_1^2 + \omega_2^2)\,\dot{z}^{\,2} + \omega_1^2\omega_2^2\, z^2\right), \qquad \omega_1 > \omega_2 > 0. \tag{6}$$

Its equation of motion factorizes:

$$\left(\frac{d^2}{dt^2}+\omega_1^2\right)\left(\frac{d^2}{dt^2}+\omega_2^2\right)z=0. \tag{7}$$

It therefore contains two ordinary frequencies, but their canonical combination has the wrong-sign structure characteristic of a ghost.

The oscillator is useful because it is finite dimensional and exactly solvable. Questions that become difficult analytic problems in a field theory reduce here to matrix algebra:

- Can the Hamiltonian be transformed to a positive normal form?
- Is that transformation unique?
- How many positive metrics exist?
- Is there a preferred one?
- What happens as $\omega_1 \to \omega_2$?
- Does the limiting Jordan system admit any positive invariant inner product?

A correct answer at the oscillator level does not automatically solve a quantum field theory. But the oscillator provides the local building block for each momentum mode of a free fourth-order field.

3 What Bender and Mannheim had already solved

Bender and Mannheim [5, 6] observed that the PU Hamiltonian should not be quantized using the naive real contour and inner product. After a complex continuation of one canonical coordinate, the Hamiltonian becomes non-Hermitian but PT-symmetric [4].

They constructed an operator

$$Q=\alpha\,pq+\beta\,xy \tag{8}$$

and the associated similarity transformation $\rho=e^{-Q/2}$ such that

$$h=\rho\,H_{\mathrm{PT}}\,\rho^{-1} \tag{9}$$

is a conventional positive Hamiltonian describing two ordinary oscillators. The original Hamiltonian is then Hermitian not with respect to the naive adjoint, but with respect to the positive metric

$$\eta=\rho^\dagger\rho=e^{-Q}, \tag{10}$$

in the general framework of pseudo-Hermitian quantum mechanics [20, 21]. The physical inner product becomes

$$\langle\psi,\phi\rangle_\eta=\langle\psi,\eta\,\phi\rangle. \tag{11}$$

With this inner product, the unequal-frequency oscillator has positive norms and a positive spectrum. Thus an important version of the ghost problem was already solved:

The free unequal-frequency Pais–Uhlenbeck oscillator admits a positive-metric quantization.

Three levels must nevertheless be distinguished, and the series addresses each in turn: the *free spectral solution* (the free oscillator maps to a positive two-oscillator Hamiltonian); *perturbative interaction stability* (the metric can be deformed away from on-shell resonances); and *global interacting viability* (energy-conserving conversion processes may obstruct the deformation and, in finite resonant systems, produce complex spectra). In short:

The free ghost problem was solved locally; Paper V determines where that solution survives interaction.

What remained unclear was the mathematical status of the construction. Why does the particular operator Q appear? Is it an ansatz or a canonical object? Is the metric unique? If it is not unique, why should the Bender–Mannheim metric be preferred? What happens when the oscillator becomes a field with infinitely many momentum modes? How does this positive construction relate to an indefinite Krein-space construction [7]? And what survives after gauge reduction in gravity? The four papers [25, 26, 27, 28] address these questions.

4 Paper I: reconstructing the Bender–Mannheim transformation

The first paper [25] reformulates the oscillator as a problem in complex symplectic geometry. Collect the canonical variables into a vector $V = (x, y, p, q)^\top$, with commutation relations encoded by the symplectic matrix J:

$$[V_a, V_b] = iJ_{ab}. \tag{12}$$

A quadratic Hamiltonian has the form $H = \frac{1}{2}V^\top GV$, where G is a symmetric matrix, and the Hamiltonian flow is generated by $A = JG$. The problem is to find a complex symplectic matrix S satisfying

$$S^\top JS = J, \qquad S^\top GS = G_0, \tag{13}$$

where G_0 is the positive two-oscillator normal form.

The first result is that, after a specified canonical normalization, this problem has a *unique* Hermitian positive-definite solution S_+. The complete solution family is the coset

$$S_+H, \qquad H \cong SO(2, \mathbb{C}) \times SO(2, \mathbb{C}), \tag{14}$$

where H is the stabilizer of the positive normal form. Thus many complex symplectic transformations diagonalize the Hamiltonian, but exactly one normalized representative is both Hermitian and positive.

The logarithm of this matrix is then converted into a quadratic quantum operator through the complexified metaplectic representation [33]. The resulting operator is precisely $Q = \alpha pq + \beta xy$ with the coefficients found by Bender and Mannheim. The logical direction has therefore been reversed:

$$\boxed{(G,\ J,\ G_0) \ \longrightarrow\ S_+ \ \longrightarrow\ Q.}$$

The operator Q is no longer introduced by guessing a useful quadratic expression. It is reconstructed from the symplectic normal-form problem.

4.1 The exact spectrum of Q

In normalized oscillator coordinates,

$$Q = r\left(a_1 a_2^\dagger + a_1^\dagger a_2\right), \qquad r = \log\frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}. \tag{15}$$

This is a number-preserving Schwinger $SU(2)$ generator. It commutes with $N_{\rm tot} = N_1 + N_2$, and on the fixed-N subspace its eigenvalues are $r(-N, -N+2, \ldots, N-2, N)$. Consequently

$$\operatorname{spec} Q = r\,\mathbb{Z}, \tag{16}$$

with pure-point spectrum and infinite multiplicities. The metric has spectrum

$$\operatorname{spec}(e^{-Q}) = \{e^{-rn} : n \in \mathbb{Z}\} \cup \{0\}, \tag{17}$$

where zero is an accumulation point rather than an eigenvalue. This provides a natural invariant domain: the algebraic Fock space of finite linear combinations of number states, preserved by Q, $e^{\pm Q/2}$, and the quadratic Hamiltonians appearing in the construction.

4.2 Many metrics, one canonical Gaussian metric

Pseudo-Hermiticity alone does not make the metric unique. If $h = \rho H_{\rm PT}\rho^{-1}$ and W is a positive operator commuting with h, then

$$\eta_W = \rho^\dagger W \rho \tag{18}$$

also defines an algebraic positive metric form [20, 21]. Thus the complete metric freedom is larger than the finite-dimensional symplectic stabilizer: the stabilizer describes the Gaussian or quadratic part of the ambiguity, while arbitrary positive functions of the normal-mode number operators provide additional non-Gaussian freedom. The first paper therefore distinguishes the *unique normalized positive symplectic diagonalizer* from the *nonunique pseudo-Hermitian metric operator.*

5 Paper II: why the canonical metric is the least-distorting one

Uniqueness inside a restricted class does not by itself explain why S_+ is geometrically natural. The second paper [26] supplies a variational characterization. For an admissible diagonalizer S, define

$$\mathcal{F}(S) = \left\|\log(S^\dagger S)\right\|_F^2. \tag{19}$$

The positive matrix $S^\dagger S$ measures how much the transformation stretches and contracts the canonical coordinates; the logarithm converts multiplicative stretching into additive "rapidities," and the Frobenius norm measures their total size. The theorem is

$$\mathcal{F}(S) \;\geq\; \mathcal{F}(S_+) = 4r^2, \tag{20}$$

with equality precisely for S_+ multiplied by a unitary element of the stabilizer — which for the PU oscillator consists of real sign matrices, so all minimizers have the same positive polar factor and determine the same Bender–Mannheim metric. In ordinary language:

$\boxed{S_+ \text{ is the least-distorting allowed transformation that makes the Hamiltonian positive.}}$

This is analogous to finding the shortest path from a point to a surface: a shortest path meets a smooth surface orthogonally. Here the "surface" is the orbit generated by transformations that preserve the positive normal form.

5.1 Cartan projection

The geometry is that of the complex symplectic group $\mathrm{Sp}(4, \mathbb{C})$ and its maximal compact subgroup $K = \mathrm{Sp}(4, \mathbb{C}) \cap U(4)$, acting on the space of positive matrices with the affine-invariant metric [34]. The Cartan projection records the logarithmic singular values of a transformation. With the doubled Gram convention used in the papers ($\mu(S) = \log \operatorname{spec}(S^\dagger S)$),

$$\mu(S_+) = (r, r), \tag{21}$$

a point on the wall $x_1 = x_2$ of the C_2 Weyl chamber. The distortion functional is related to the Cartan norm by

$$\mathcal{F}(S) = 2\|\mu(S)\|^2. \tag{22}$$

Thus the minimum-distortion theorem is a Cartan-projection theorem.

5.2 Why the theorem is not a routine Kempf–Ness application

The physical stabilizer $H \cong SO(2, \mathbb{C})^2$ is not compatible with the canonical Cartan involution determined by the physical inner product: generically

$$H \cap K \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \tag{23}$$

rather than the maximal compact $U(1)^2$ of the abstract group. The obstruction belongs to the *embedding* of H, not to its abstract group structure, so the classical nearest-point theory of self-adjoint groups [35, 36] does not apply to H directly.

The resolution is to enlarge H to its smallest compatible (θ-stable) hull. Generically this is

$$SL(2, \mathbb{C}) \times SL(2, \mathbb{C}), \tag{24}$$

the mode-preserving subgroup, whose positive part is a totally geodesic copy of $\mathbb{H}^3 \times \mathbb{H}^3$. The Bender–Mannheim direction is orthogonal to this mode-preserving geometry, so standard nonpositively curved projection theory can be applied to the hull and the result restricted back to the physical stabilizer. The paper proves a general principle:

> *A positive diagonalizer minimizes Cartan distortion along a compatible subgroup exactly when its Hermitian generator is orthogonal to the subgroup's positive tangent directions.*

For a splitting into many oscillator modes, those tangent directions are the mode-preserving deformations, and the normal directions are precisely the inter-mode couplings. The PU oscillator is the simplest nontrivial example: its preferred generator is entirely inter-mode.

6 The exceptional point: when two frequencies merge

The rapidity is $r = \log \frac{\omega_1+\omega_2}{\omega_1-\omega_2}$. As $\omega_1 \to \omega_2$, the denominator tends to zero and $r \to \infty$. The positive diagonalizer stretches one pair of directions by $e^{r/2}$ and contracts another by $e^{-r/2}$; the corresponding Cartan distance diverges.

This is not merely a poor choice of coordinates. At equal frequency, the dynamical matrix ceases to be diagonalizable and develops a Jordan block — the hallmark of an exceptional point [38, 39]. A 2×2 Jordan block has the form

$$H_J = \begin{pmatrix} \omega & 1 \\ 0 & \omega \end{pmatrix}. \tag{25}$$

Suppose there were a positive-definite Hermitian form η satisfying $H_J^\dagger \eta = \eta H_J$. Then the matrix $\eta^{1/2} H_J \eta^{-1/2}$ would be Hermitian, hence diagonalizable. But similarity transformations preserve Jordan form. This is impossible. Hence a nontrivial Jordan block admits no nondegenerate positive-definite invariant Hermitian form [25, 37]. It may still admit:

- degenerate positive-semidefinite forms;
- nondegenerate indefinite forms;
- or intrinsically Jordan PT-symmetric realizations [6].

What fails is the continuation of the unequal-frequency positive Hilbert metric as a nondegenerate invariant positive form. This distinction matters: the conclusion is not that the equal-frequency theory cannot exist, but that it cannot remain an ordinary diagonalizable positive-metric system of the same kind.

7 From one oscillator to a field

Consider the free fourth-order scalar field

$$(\Box + m_1^2)(\Box + m_2^2)\, \phi = 0. \tag{26}$$

After Fourier transforming in space, each spatial momentum k contributes a PU oscillator with frequencies $\omega_i(k) = \sqrt{k^2 + m_i^2}$. The modewise rapidity is

$$r(k) = \log \frac{\big(\omega_1(k) + \omega_2(k)\big)^2}{m_1^2 - m_2^2} = 2 \log |k| + O(1) \qquad (|k| \to \infty). \tag{27}$$

Thus the modewise similarity transformations become increasingly nonuniform in the ultraviolet.

This is where finite-dimensional and field-theoretic questions diverge. Each individual momentum mode can be treated successfully, while the collection of infinitely many modes may define an unbounded field transformation, a non-equivalent norm on phase space, a different quasifree representation, or a different real form of the field algebra. A modewise solution is therefore necessary but not sufficient for a field theory.

8 Metric geometry and vacuum geometry are different

A central lesson of the field papers [26, 27] is

$$\boxed{\text{metric distortion} \neq \text{vacuum excitation.}}$$

The Cartan rapidity $r(k)$ diverges like $2\log k$, and the finite-dimensional observable-space metric has eigenvalues $e^{\pm r(k)} \sim k^{\pm 2}$. This describes a strong deformation of the canonical phase-space norm. It does *not* follow that the transformation creates $O(k^2)$ particles.

In normalized coordinates the generator is number preserving:

$$Q_k = r(k)\big(a_{1,k}a_{2,k}^{\dagger} + a_{1,k}^{\dagger}a_{2,k}\big). \tag{28}$$

It acts inside the complexified beam-splitter algebra, and the canonical vacuum is annihilated by it, even though the associated Cartan norm is arbitrarily large.

The reason is geometric. Positive metrics are points in one quotient space, roughly G/K, while Gaussian vacuum rays belong to a different quotient, G/P_Ω, where P_Ω stabilizes the vacuum line. The vacuum-ray map does not factor through the metric quotient. Two transformations can therefore have very different metric distortion while producing the same vacuum ray. This prevents one from estimating physical particle content directly from the Cartan norm.

9 The universal complex vacuum covariance

For the fourth-order field, the selected two-point function has the spectral form

$$\widetilde{W}_{12}^{+}(p) = \varepsilon\,\theta(p^0)\,\frac{\delta(p^2 - m_1^2) - \delta(p^2 - m_2^2)}{m_1^2 - m_2^2}, \tag{29}$$

where $\varepsilon = \pm 1$ records the overall action convention. This is a divided difference of two ordinary Klein–Gordon two-point functions:

$$W_{12}^{+} = \varepsilon\,\frac{W_{m_1}^{+} - W_{m_2}^{+}}{m_1^2 - m_2^2}. \tag{30}$$

The field paper [27] shows that the admissible positive metrics all select the same complex covariance. They may change the physical adjoint and the representation of observables, but not this underlying spectral two-point distribution. The result should be stated carefully:

The complex covariance is metric-independent; the involution and completion need not be.

A state in the strict algebraic sense requires a fixed involution and a positivity condition. When different completions use different involutions, it is safer to speak first of the common complex quasifree covariance.

10 Resonance: the metric diverges but the covariance survives

Let $m_1^2, m_2^2 \to m^2$. Then the divided difference has the regular limit

$$\widetilde{W}^{+}_{mm}(p) = -\varepsilon\, \theta(p^0)\, \delta'(p^2 - m^2). \tag{31}$$

In the time domain, per mode,

$$W^{+}_{mm}(t, k) = -\varepsilon\, \frac{(1 + i\omega t)\, e^{-i\omega t}}{4\omega^3}, \qquad \omega = \sqrt{k^2 + m^2}. \tag{32}$$

The positive diagonalizing transformation diverges because the dynamics becomes Jordan. Yet the complex vacuum covariance remains finite. This is one of the main conceptual results:

Resonance is a singularity of the metric and dynamics, not necessarily of the vacuum covariance.

At the doubly massless point $m = 0$, the covariance is related, after matching the action sign and infrared convention, to the massless $\Box^2$ covariance used in the Krein-space construction of Bateman and Turok [7]. Their sign convention gives the opposite overall covariance to one common PU convention: the relation is an action-sign reversal, not a literal equality before conventions are matched. This bridge identifies the *free complex covariance*; it does not by itself reconstruct Bateman and Turok's interacting $O(1, 1)$ embedding, ghost symmetry, or indefinite-space probability rule, which are additional structures of their theory.

11 What is a Krein space?

A Hilbert space has a positive inner product: $\langle \psi, \psi \rangle > 0$ for every nonzero state. A Krein space has a nondegenerate but *indefinite* inner product: a state can have positive, negative, or zero norm.

At first sight this appears incompatible with probability. But indefinite-metric quantum theories can possess an additional grading or symmetry that identifies the physical probability structure. In the Bateman–Turok construction [7], a hidden ghost-parity symmetry plays this role.

The positive and Krein descriptions sacrifice different conventional requirements. For an unpaired ghost oscillator, one may try to demand:

1. positive norm;
2. positive energy;
3. standard reality of the original field.

The three cannot all be retained simultaneously [28]. A complex quarter-turn of the ghost variables can make the energy and norm positive, but the original amplitude becomes anti-Hermitian; i times the amplitude is the physical Hermitian observable. Alternatively, the original field can remain real and the energy spectral condition can be retained, but the state space becomes indefinite. This gives two real forms of a common complex system:

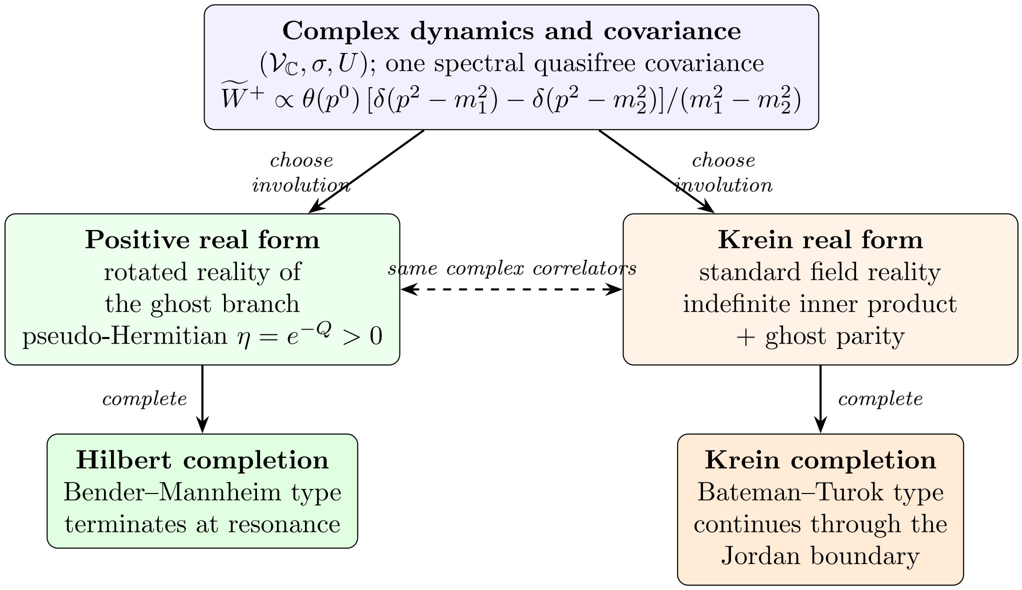

Figure 1: The three-level structure: one complex spectral covariance, two covariant real forms (involutions), two completions. The two branches agree on all complex correlation functions and differ in which operators are Hermitian and how probabilities are represented. At the resonant (Jordan) boundary only the Krein branch continues as a nondegenerate completion.

	positive completion	Krein completion
energy	positive	positive
norm	positive	indefinite
original field reality	rotated	standard

The complex equations and complex covariance may agree even though the physical adjoints differ. Figure 1 summarizes the resulting three-level structure.

12 A necessary representation-theoretic distinction

There are two different ways to compare the positive PT construction with an ordinary oscillator Fock space, and conflating them produces contradictions [26].

12.1 Physical Dyson transport

For each mode, the PT ground state is exactly the Dyson pullback of the positive normal-form vacuum,

$$\psi_k = \rho_k^{-1}\phi_k. \tag{33}$$

Equip the PT space with the metric $\eta_k = \rho_k^\dagger \rho_k$. Then

$$\rho_k : \ \mathcal{H}_{\mathrm{PT},k} \longrightarrow \mathcal{H}_{\mathrm{normal},k} \tag{34}$$

is unitary with respect to the physical inner products, and $\rho_k\psi_k = \phi_k$ *exactly.* The pointed modewise unitaries therefore tensor together without any vacuum-overlap obstruction: the completed positive PT theory is globally equivalent to its positive normal-form theory.

12.2 Identity embedding in an auxiliary standard CCR algebra

One may instead place both Gaussian wavefunctions in the same standard Schrödinger L^2 space, retain the same canonical variables and standard adjoint, and compare the vectors through the identity embedding. Under that comparison, the per-mode Gaussian fidelity approaches $\sqrt{3}/2$, the expectation of the standard auxiliary number operator approaches $1/3$, and the infinite products of these auxiliary standard-CCR states are disjoint [40, 26].

This does not contradict the physical Dyson equivalence: the two comparisons identify observables differently.

$$\boxed{\text{identity embedding of auxiliary variables} \neq \text{Dyson transport of the physical algebra.}}$$

This is an example of why field representation theory cannot be discussed without specifying the algebra, its involution, the embedding between descriptions, and the chosen completion.

13 The ultraviolet Fock-sector hierarchy

The preceding comparison placed a selected PU vacuum beside the split two-field vacuum. Paper III also asks a different question: when do two *selected* vacua, belonging to different mass pairs (m_1, m_2) and $(\bar{m}_1, \bar{m}_2)$, define the same ultraviolet Fock sector in common field variables? Writing $\Sigma = m_1^2 + m_2^2$ and $\Pi = m_1^2 m_2^2$, the modewise fidelity obeys

$$1 - f(k) = \frac{(\Sigma - \bar{\Sigma})^2}{12k^4} + O(k^{-6}). \tag{35}$$

If $\Sigma = \bar{\Sigma}$, the k^{-6} term also vanishes and the first remaining obstruction is proportional to $29(\Pi - \bar{\Pi})^2/(576k^8)$. Since the two symmetric invariants (Σ, Π) determine the unordered mass pair, the ultraviolet classification terminates:

spatial dimension	ultraviolet sector invariant
$d \leq 3$	none
$4 \leq d < 8$	$\Sigma = m_1^2 + m_2^2$
$d \geq 8$	(Σ, Π), the unordered mass pair

At $d = 4$ and $d = 8$ the corresponding obstruction is logarithmic, so the endpoints belong to the higher-dimensional rows shown.

This is an *ultraviolet* classification, not automatically a global GNS-equivalence theorem. For strictly positive masses the infrared is regular. Against the doubly massless $\Box^2$ vacuum, however, the relative occupation behaves as k^{-3} and its infrared integral diverges for $d \leq 3$. Whether the massless anchor belongs to the same global sector in those dimensions therefore remains open. This is why "one ultraviolet sector" must not be strengthened to "one global representation."

14 The fourth-order vacuum is locally well behaved

Ordinary four-dimensional Klein–Gordon two-point functions have a leading short-distance singularity proportional to $1/\sigma_+$, where $\sigma_+(x, y)$ is the regularized squared geodesic separation.

In the fourth-order divided difference, this mass-independent leading term cancels. The result has the Hadamard-like form

$$W_{12}^{+}(x,y) = V_{12}(x,y)\,\log\sigma_{+}(x,y) + H_{12}(x,y), \tag{36}$$

where V_{12} and H_{12} are smooth and

$$V_{12}(x,x) = \frac{\varepsilon}{16\pi^2}. \tag{37}$$

There are further terms such as $\sigma\log\sigma$, so the two-point function is not simply a constant logarithm plus a smooth remainder. Nevertheless, its wavefront directions are the standard positive-frequency Hadamard directions, in the microlocal sense of [41].

This is the natural fourth-order analogue of the Hadamard property [27]. It says that the global ghost or completion problem does not automatically produce a pathological local ultraviolet singularity: the vacuum can be globally unusual in its real form or representation, yet locally well behaved in the microlocal sense required for renormalized observables [42, 43].

15 When interactions are switched on

Everything so far concerns free theories. The fifth paper of the series [29] asks the next question: can the free positive or Krein structure be continuously deformed when nonlinear processes are switched on? The answer has four parts: generically yes at low perturbative order away from resonances; no on specific energy-conserving conversion channels; in field theory continuous momentum makes such channels kinematically available for generic split masses, with the obstruction verified on a nonempty open shell subset; and an exact doubled sector-exchange symmetry can survive even though the analytic deformation of the canonical positive pointed-sector metric fails there. Figure 2 summarizes the mechanism.

15.1 Deforming the positive metric

Perturb the free theory, $H_\zeta = H_0 + \zeta V$, and recall the free Dyson map with $\rho_0 H_0 \rho_0^{-1} = h_0 = h_0^\dagger$. One seeks a corrected map

$$\rho_\zeta = e^{-\frac{1}{2}(\zeta R_1 + \zeta^2 R_2 + \cdots)}\rho_0, \tag{38}$$

keeping the transformed Hamiltonian Hermitian order by order. At first order,

$$[h_0, R_1] = v^\dagger - v, \qquad v = \rho_0 V \rho_0^{-1}. \tag{39}$$

The interaction introduces a new "bad" non-Hermitian part, and the correction R_1 asks whether the measuring geometry can be adjusted to absorb it. The key principle is simple: a correction can remove terms that move states between *different* energies; it cannot remove a term that acts entirely among states with exactly the *same* energy. When two states have different energies, the metric can distinguish them; when they have exactly equal energy, the interaction is resonant and cannot be rotated away. Technically, the obstruction is the energy-conserving part of the source,

$$\mathfrak{o}_+ = \Pi_{\ker \mathrm{ad}_{h_0}} S, \tag{40}$$

the projection onto processes satisfying exact free-energy conservation.

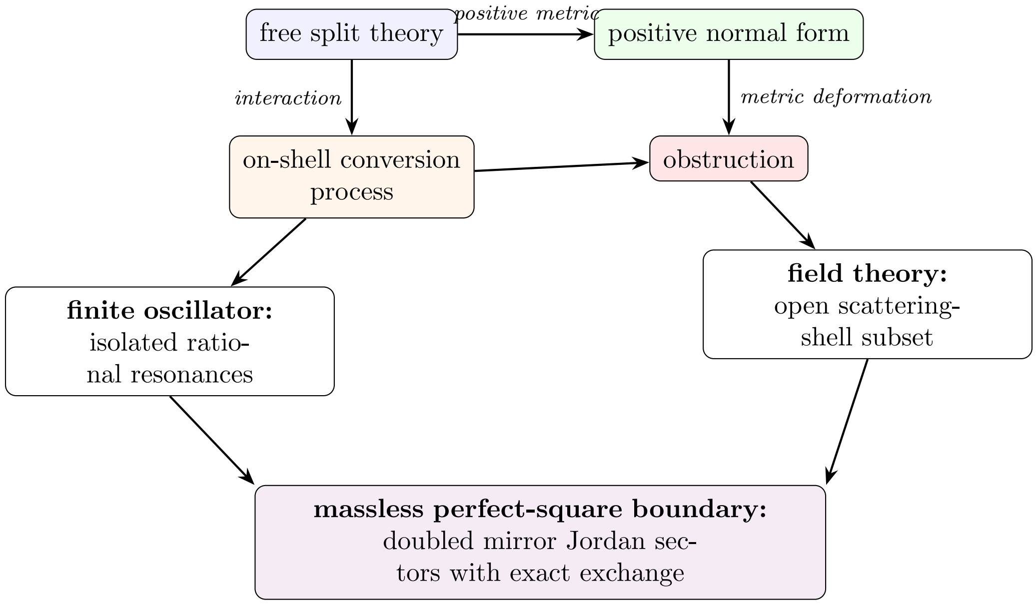

Figure 2: The interaction mechanism of [29]: metric deformation meets on-shell conversion; isolated resonances in the oscillator become kinematically available scattering shells in the field theory, with the obstruction verified on a nonempty open subset; at the massless boundary a doubled exchange structure survives.

15.2 A finite oscillator fails only at special resonances

For the cubic PU interaction at generic unequal frequencies, the positive metric can be corrected through the computed orders — the explicit generators exist and the transformed Hamiltonian is Hermitian. But at

$$\omega_1 = 3\omega_2, \tag{41}$$

one high-frequency quantum can convert into three low-frequency quanta without changing the total energy, and the second-order correction is obstructed by exactly that conversion operator, $a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3$. At $2\omega_1 = 3\omega_2$ a third-order obstruction appears. A transfer-lattice rule organizes which conversion monomials can first occur at each order. Its first new prediction has been confirmed computationally: at $\omega_1 = 5\omega_2$ the second- and third-order classes vanish, while the fourth-order class is a nonzero $1 \leftrightarrow 5$ conversion proportional to $a_1 a_2^{\dagger 5} - a_1^\dagger a_2^5$. This confirms one prediction of the proposed hierarchy; the general $p{:}q$ rule remains a conjecture.

Imagine two musical notes: usually they are unrelated, but if one note has exactly three times the frequency of the other, one vibration can split into three without changing the total energy — at that point the interaction is perfectly synchronized with the system.

The 3:1 obstruction is more than a failure of one formula. Sufficiently excited resonant multiplets develop complex-conjugate energies, $E_\pm = E_R \pm i\Gamma$, at second order. This is formally defined perturbative PT breaking: [29] constructs the antilinear symmetry explicitly, proves that it commutes with the interacting Hamiltonian, and certifies the complex pair *exactly* in the effective resonant multiplet (by rational algebra and Sturm counting, not only numerically). Where the complex pair persists, no positive-definite invariant metric can make the Hamiltonian equivalent to an ordinary self-adjoint one. Two levels of conclusion must not be merged: this complex-spectrum statement is established in the verified resonant oscillator setting; the field-theory result below is presently a perturbative metric obstruction, not yet a demonstrated complex spectrum.

15.3 The momentum continuum turns exceptional resonances into ordinary scattering

For an oscillator, the frequencies are fixed numbers and exact resonances require special ratios. For a field, $E_a(k) = \sqrt{k^2 + m_a^2}$, and momenta vary continuously: one can satisfy

$$E_H(k_1) + E_L(k_2) = E_L(k_3) + E_L(k_4) \tag{42}$$

together with momentum conservation on open sets of scattering data, for *any* split masses. The key process is the branch-changing scattering $H + L \to L + L$. Paper V shows that the second-order positive-metric source is nonzero on an open scattering-shell subset (exactly, at a rational kinematic point, hence on an open neighborhood — not claimed at every kinematic point): the continuum turns the oscillator's isolated resonance obstruction into an obstruction on a nonempty open part of a scattering shell. In a small mechanical system, resonance requires the machine to be tuned to a special ratio; in a field, particles can change their momentum, and even if their masses are not specially tuned, their motion can supply the exact energy match.

The canonical positive completion therefore remains meaningful locally and perturbatively away from resonant processes, but it cannot be deformed analytically into a positive pointed-sector metric on the verified branch-changing shell subset. One qualification is essential: this establishes that *no analytic deformation of the canonical metric* exists there — not that no conceivable positive completion of any kind exists. For the oscillator with certified complex energies the stronger conclusion is available; for the field theory the perturbative statement is what is proved.

15.4 Why the obstruction waits until second order

The perfect-square cubic vertex is exactly the Källén polynomial, $\mathcal{V}_3 = \frac{1}{2}\lambda_{\mathrm{K}}(p_1^2, p_2^2, p_3^2)$ (Section 14 notation). It vanishes for three ordinary massless legs and has a second-order zero at the confluent massless point; and the transported reality factors eliminate the first-order positive obstruction on the physically open decay channel. So the positive metric survives the first test. But at second order, two cubic interactions combine into a branch-changing $2 \to 2$ process, and the obstruction appears: a single interaction event is too constrained to cause the problem, while two interaction events in sequence can convert the particles while preserving total energy — and that is where the positive geometry fails.

15.5 A different structure survives: two mirror Jordan sectors

The perfect-square theory admits an exact rewriting, introduced by Bateman and Turok [7], who also identified its exchange symmetry as ghost parity and as charge conjugation for an internal $O(1,1)$ symmetry; the sector-level reading given here is what the interaction paper adds. Introducing an auxiliary field and the nonlinear change of variables $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$,

$$\mathcal{L} = -\partial_\mu U\, \partial^\mu V + \frac{\lambda^2}{2} U^2 V^2. \tag{43}$$

The constant zero-potential stationary backgrounds form two branches, $(U_0, 0)$ and $(0, V_0)$, and the exact exchange $U \leftrightarrow V$ maps one branch to the other. Around $(1, 0)$ the linearized equations are $\Box v = 0$, $\Box u = \lambda^2 v$ — a Jordan chain generated by the interaction itself; around $(0, 1)$ the chain has the opposite orientation. The complete theory thus has two mirror valleys: choosing one valley gives one orientation of the Jordan system, and the exact symmetry does not rearrange particles inside that valley — it carries the entire valley to its mirror.

Three distinctions matter here. The exact symmetry belongs globally to the *extended* (U, V) theory; its expression in the original (ϕ, ψ) variables is a chart-transition map, singular at the chosen perturbative vacuum; and it is not a ghost-number sign inside one Fock sector. It may ultimately form part of an antiunitary CPT-like structure, but what is proved is a linear sector exchange, not a quantum CPT theorem.

Finally, the ordinary split-theory ghost parity becomes singular at the merge. In the confluent Jordan basis $c = (b_+ + b_-)/2$, $d = (b_+ - b_-)/\delta$, the branch parity becomes

$$P_\delta = \begin{pmatrix} 0 & 2/\delta \\ \delta/2 & 0 \end{pmatrix}, \qquad P_\delta^2 = 1, \tag{44}$$

with no bounded limit on a single Jordan chain: as the two particle types merge, the old rule "one is even and one is odd" becomes infinitely badly conditioned. The only finite remnant is to exchange the entire Jordan system with its mirror copy: after doubling by the oppositely oriented sector, the singular parity becomes a finite sector exchange. This confluent result is the *linear bridge* to the independently exact nonlinear $U \leftrightarrow V$ symmetry: what the regulated parity converges to is the linearization of that exchange around the two mirror backgrounds — no stronger identification is claimed.

The relation between the two structures is sharp in the verified perturbative setting. At the rational shell point used to certify the field obstruction, the complete reachable second-order on-shell matrix is *exactly* pseudo-Hermitian with respect to ghost-branch parity, and its entire positive-metric obstruction lies in the parity-odd block. On the same matrix elements the Krein pairing is preserved while the analytic deformation of the canonical pointed positive metric is obstructed. The two escapes from the ghost problem therefore *separate*: the Krein completion is not a hidden version of that positive completion. This does not exclude a nonanalytic or differently pointed positive construction, nor does it prove a complex field-theory spectrum.

There is a further boundary distinction. Exact pseudo-Hermiticity of the split on-shell matrix is weaker than the one-sided charge condition used in the Bateman–Turok generalized Born rule. A canonical map to their cross-paired charge basis sends the regulated pointed vacuum to a Gaussian with

$$\frac{S_{UU}}{S_{VV}} = \left(\frac{\delta}{2g}\right)^2 = \frac{\epsilon}{g}, \qquad \delta^2 = 4\epsilon g. \tag{45}$$

Thus both charge signs are intrinsic at split mass. The image becomes one-sided only on the $O(1, 1)$-symmetric confluent line $\epsilon = 0$; the Bateman–Turok massless point additionally has $\mu^2 = 0$. A self-contained charge-null lemma then makes every strictly negative component null and orthogonal to the neutral component. The remaining capstone calculation is to transport the explicit obstruction through this boundary limit and evaluate its generalized Born trace.

Terminology.

Positive pseudo-Hermitian metric: a positive inner product obtained by changing the adjoint through a Dyson map. *Metric deformation*: an order-by-order correction of that inner product after an interaction is introduced. *Obstruction*: the energy-conserving part of the unwanted non-Hermitian interaction that cannot be generated by a metric correction. *Pointed sector*: a quantum construction built around one selected vacuum background. *Doubled theory*: a completion containing both mirror Jordan sectors. *Sector-exchange involution*: the exact map exchanging the two mirror backgrounds and reversing the Jordan-chain orientation. *Spectral breaking*: the appearance of complex-conjugate energy levels, which excludes a positive invariant metric where the complex spectrum persists. "Ghost parity" is used only with a qualifier: split branch-number parity, the exact $U \leftrightarrow V$ exchange, or a (future) antiunitary CPT-like operation are different objects.

16 Why gravity is not merely five copies of the oscillator

The scalar PU oscillator has no gauge symmetry. Gravity does.

Consider linearized scalar-free Einstein–Weyl gravity around flat spacetime: the Einstein term plus a tuned quadratic-curvature term ($\alpha R_{\mu\nu}^2 + \beta R^2$ with $\alpha = -3\beta$). The linearized spectrum contains a massless graviton with two helicities and a massive spin-2 branch with five helicities [24]. A naive polarization-by-polarization treatment might suggest that every massive helicity belongs to an independent PU pair. Gauge reduction shows otherwise [28]: the reduced phase space has the structure

$$\Gamma_{\text{phys}} \cong 2\,\Gamma_{\text{PU}} \;\oplus\; 2\,\Gamma_{-} \;\oplus\; \Gamma_{-}. \tag{46}$$

The two helicity-2 sectors are genuine fourth-order PU pairs: each contains a massless graviton mode and a massive ghost partner. The helicity-1 and helicity-0 sectors contain only *unpaired* second-order massive ghosts: their apparent massless partners lie entirely in the diffeomorphism gauge orbit and disappear under reduction.

Gauge reduction preserves PU pairing only in helicity ± 2.

This is the first major way in which gravity differs from the scalar model.

17 Lorentz covariance forbids hybrid solutions

The five massive helicities $+2, +1, 0, -1, -2$ form one irreducible massive spin-2 representation. They are not five independent species that can be quantized separately: a Lorentz transformation can mix them.

By Schur's lemma, an invariant Hermitian form on the irreducible spin-2 polarization space is proportional to the identity. Its sign or reality assignment must therefore be uniform across all five helicities. This rules out a tempting hybrid construction in which the helicity-2 modes receive the positive PU metric while the helicity-1 and helicity-0 ghosts are treated in a Krein space: such an assignment is not Lorentz covariant.

Within the class analyzed in the gravity paper [28] — free, translation-invariant, quasifree, mode-local constructions compatible with the reduced symplectic form, the spectral condition, and Poincaré covariance — there are exactly two covariant completion classes:

1. a positive pseudo-Hermitian completion in which the full massive spin-2 multiplet has uniformly rotated reality;
2. a Krein completion in which the gravitational field retains standard reality and the entire massive spin-2 multiplet carries a uniform negative sign relative to the positive massless graviton sector.

The two completions induce the same complex gauge-invariant covariance, while differing in their involution and completion. This is the gravitational version of the real-form distinction of Figure 1.

18 The gauge-invariant gravitational bridge

The equality of the two complex covariances is not merely a polarization-by-polarization analogy. The reduced helicity kernels have residues fixed by the reduced symplectic form. Their ratios $1 : M^2 : \frac{1}{3}$ match the gauge-invariant contractions of the massive spin-2 projector,

$$\frac{1}{2} : \frac{M^2}{2} : \frac{1}{6} \tag{47}$$

so they reassemble, modulo pure-gauge bi-distributions, with one overall coefficient $4/c_1$. The massless shell couples only to the transverse-traceless polarizations. Thus both real forms determine the same complex spectral functional on the algebra of linear gauge-invariant observables, rather than merely agreeing in a chosen helicity frame.

The massive projector itself contains apparently singular terms $p_\mu p_\nu / M^2$. Applying the linearized Weyl-tensor map annihilates all of them: the Weyl–Weyl kernel is gauge independent, covariant, and regular as $M \to 0$, with the standard positive-frequency Hadamard wavefront directions. This cancellation is why the conformal limit below is formulated on the curvature-generated regular subalgebra. The raw scalar potential kernel does not have a regular limit; the divergent direction becomes Weyl-null and must first be quotiented out.

Relation to CPT language. Mannheim's CPT-based positive scalar product belongs, away from the conformal locus, to the positive class just described, with its uniformly rotated massive reality [23]. At the Jordan boundary the no-go theorem excludes a time-independent, nondegenerate, positive-definite invariant form; it does not exclude the intrinsically Jordan realization of Bender and Mannheim, which uses nonstationary Jordan states and a degenerate pairing [6]. The two statements are therefore compatible: the theorem delimits which positive completion terminates, rather than declaring the equal-frequency theory nonexistent.

19 The conformal Jordan boundary

Let the coefficient of the Einstein term tend to zero while the quadratic-curvature coupling remains fixed. The massive spin-2 mass then tends to zero, and the different helicity sectors behave differently [28].

Helicity ± 2. The massless and massive branches merge, producing two $\Box^2$ Jordan systems. Each polarization contributes two configuration-space solutions, giving four modes in total.

Helicity ± 1**.** The vector ghosts have an ordinary massless limit. They remain second-order modes rather than becoming Jordan partners.

Helicity 0**.** The scalar kinetic normalization tends to zero. At the conformal point a new Weyl gauge symmetry appears, $\delta h_{\mu\nu} = 2\sigma\eta_{\mu\nu}$: the scalar direction becomes null in the presymplectic form and is removed by the quotient.

The conformal count is therefore $4+2+0=6$. The positive split-frequency metric runs to infinite Cartan distance and has no nondegenerate positive continuation through the Jordan boundary; the complex spectral covariance and the standard-reality Krein form can continue, provided the newly null Weyl direction is quotiented out. The algebra of observables changes rank at the boundary, so convergence must be formulated on a fixed regular family of observables — most naturally through gauge- and Weyl-invariant curvature quantities. (Boundary constructions that instead *remove* one branch of the solution space, as in [44], and the richer phase diagram at nonzero cosmological constant [45, 46], are related but distinct stories.)

20 Paper VI: interaction obstruction in Einstein–Weyl gravity

Paper VI [30] asks the interacting question left open by the free gravity classification. Scalar-free Einstein–Weyl gravity around flat space contains a healthy massless graviton h and a ghost-signed massive spin-2 multiplet M. Paper IV found two covariant free real-form classes: a positive completion obtained by uniformly quarter-turning the complete massive multiplet, and a standard-reality Krein completion with uniformly negative massive signature. Paper V supplied the diagnostic. If ρ_0 is the free Dyson map, an analytic interacting deformation must solve cohomological equations

$$[h_0, R_n] = S_n, \tag{48}$$

so the projection of S_n onto exact free-energy degeneracies must vanish.

Gravity is protected at first order for two independent reasons. The Einstein solution space is an exact nonlinear subsector of Einstein–Weyl gravity, so tree amplitudes with exactly one massive external field and otherwise massless gravitons vanish. The remaining odd-massive cubic vertex MMM is nonzero on complex kinematics but has no real equal-mass $1 \leftrightarrow 2$ shell. Consequently the first-order positive-metric obstruction vanishes.

At second order the process

$$M + M \longrightarrow M + h \tag{49}$$

is physically open. Paper VI establishes its nonvanishing by two independent calculations. An Einstein-frame factorization calculation produces a nonzero massive-pole residue. A separate calculation in the original fourth-order variables includes the quartic contact term, all s-, t-, and u-channel exchanges, physical polarization and equation-of-motion conditions, the reverse-process adjoint, Ward identities, and two internal gauge representatives. At an interior rational point of the real physical shell it gives a nonzero exact amplitude.

The essential bridge is not inferred from that amplitude's phase alone. In a regulated stationary Fock space, define the second Born operator

$$B_{2,+}(E) = P_E\big(v_2 + v_1 Q_E (E-h_0)^{-1} Q_E v_1\big) P_E. \tag{50}$$

With the spectral solution for R_1, Paper VI proves the identity

$$P_E \left[(v_2^\dagger - v_2) + \frac{1}{2}[R_1, v_1 + v_1^\dagger] \right] P_E = B_{2,+}(E)^\dagger - B_{2,+}(E). \tag{51}$$

The covariant contact-plus-exchange calculation matches this stationary Born operator after the usual propagator decomposition, external residue normalization, and identical-particle factors. The calculation therefore evaluates the complete deformation source, not merely a phase-sensitive scattering amplitude.

Covariance quarter-turns the complete massive multiplet. Counting the massive fields at all vertices and the compensating signs of internal massive propagators shows that every connected diagram with E_M external massive legs acquires the total phase $(-i)^{E_M}$. The process $MM \to Mh$ has $E_M = 3$; its nonzero real standard-frame Born block thus becomes anti-Hermitian in the positive frame. Equation (51) then gives a nonzero second-order shell cocycle. Hence no analytic second-order deformation based at the canonical Paper-IV positive real form exists in the stated tree-level perturbative class on a nonempty open subset of this shell.

The Krein result is different. Agreement with the free gravitational signature and natural particle-number-diagonal, cluster-multiplicative second quantization gives the conventional lift

$$J_F = (-1)^{N_M} \tag{52}$$

on physical BRST cohomology. The process crosses its positive and negative sectors, but oddness under a $\mathbb{Z}_2$ grading does not imply nullity: the obstruction has a nonzero quadratic Krein trace. Moreover, the verified nonzero MMM and MMh vertices force every uniform abelian charge on the irreducible massive multiplet, with neutral graviton, to be trivial. The scalar boundary's one-sided $O(1,1)$ charge mechanism therefore has no direct gravitational analogue.

This conclusion is deliberately limited. Paper VI excludes an analytic second-order deformation of the canonical positive free metric and a null interpretation under the conventional multiplicative massive-number grading. It does not exclude indefinite Krein evolution, nonanalytic or differently pointed positive metrics, momentum-dependent or non-factorizing involutions, singular conformal-boundary limits, or boundary conditions that truncate part of the fourth-order solution space.

21 Papers VII–VIII: residual states and covariant transport

Papers VII–VIII [31, 32] return to the pure-Weyl boundary of Paper IV, but they answer two different questions. Paper VII asks what survives the residual gauge quotient after a state polarization has been chosen. Paper VIII asks whether that state-side answer is computed by the covariant compactly supported metric BV complex. The second theorem imports the first; it does not repeat its Chevalley–Eilenberg calculation.

21.1 The selected closed-universe problem

Both papers work on

$$M = \mathbb{R} \times S^3 \tag{53}$$

and make the same explicit boundary choice. The spatial slice is closed, there is no asymptotic charge or external clock, the cylinder is not deparametrized, and all fifteen conformal reducibilities—including compact time translation D—are gauged. This is not the fixed-cylinder problem in which D is retained as a Hamiltonian.

The charge audit makes this restriction mathematically visible. On the unrestricted locally reduced linearized space $\mathcal{P}_{\mathrm{lin}}$, D has the nonzero integrable kernel $M_D = -\frac{1}{2} J_{\mathrm{conf}} D$ and is a physical Hamiltonian symmetry. On the common Taub zero fibre $\mathcal{P}_{\mathrm{Taub0}}$, however, the inclusion ι obeys

$$\iota^* \iota_{X_D} \Omega_\Sigma = d(\iota^* \mu_D) = 0. \tag{54}$$

Thus the compact verdict is sector-dependent: the quotient used below is legitimate only after the explicit full moment-map zero restriction. The cross-programme validation dossier keeps this classical statement separate from the still-open clock, boundary, interacting, and quantum questions.

Three reductions remain conceptually distinct:

$$\begin{aligned} \text{covariant metric BV complex} &\longrightarrow \text{Cauchy/energy one-particle complex} \\ &\longrightarrow \text{polarized residual state complex.} \end{aligned} \tag{55}$$

Paper VIII proves the first comparison and its compatibility with the second. Paper VII performs the final residual reduction. A local propagating mode can therefore be present in the middle complex while contributing no one-particle class to the last one.

21.2 Paper VII: the state-side theorem

The locally reduced one-particle coefficient module is the parity-complete on-shell Weyl module $\mathcal{W}_+ \oplus \mathcal{W}_-$. Its invariant form has the branch signs

$$J_{\mathrm{conf}} = +I_E \oplus (-I_A) \oplus (-I_L). \tag{56}$$

After the selected positive-frequency Lagrangian polarization, the coefficient algebra is

$$\mathcal{F}_{\mathcal{W}} = \mathrm{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-), \tag{57}$$

and the residual ghosts give the absolute $\mathfrak{so}(4,2)$ Chevalley–Eilenberg complex.

Theorem 21.1 (Paper VII: selected residual state theorem)**.** *For the selected formal closed-universe BV–BFV problem, the Taub constraints are the residual moment-map components, endpoint duality and the normalized BV–BFV suspension supply exactly one residual cotangent pair, and the positive-frequency state complex is*

$$\mathcal{H}_{\mathrm{state}} = \mathcal{F}_{\mathcal{W}} \otimes \Lambda^\bullet \mathfrak{so}(4,2)^*. \tag{58}$$

Its centered one-particle group vanishes, while

$$\boxed{H^4_{\mathrm{res}} = \mathrm{span}\{[W_+^2], [W_-^2]\}, \qquad \mathrm{sig}\, G_{\mathrm{res}} = (2,0), \qquad G_{\mathrm{res}} = I_2.} \tag{59}$$

The same result holds in the selected energy-mode Krein–Fock completion.

The classes in (59) are ghost-dressed deformation or vertex classes. They are not two one-particle gravitons. Their parity combinations descend to the Weyl-square and Pontryagin directions; after the local Euler–Lagrange quotient only the Weyl-square direction is dynamical.

21.3 Paper VIII: the covariant-side theorem

Paper VIII retains the metric variables and adds curvature only through a support-local prolongation. Its analytic task is a chain contraction, not a second residual cohomology computation.

Theorem 21.2 (Paper VIII: covariant causal transport)**.** *The complete free pure-Weyl metric BV complex on the conformal cylinder admits a support-local curvature prolongation and retarded/advanced degree-(-1) maps $\Lambda_\pm$ such that*

$$Q\Lambda_\pm + \Lambda_\pm Q = 1, \qquad \operatorname{supp}(\Lambda_\pm f) \subseteq J^\pm(\operatorname{supp} f), \qquad \Lambda_+^\sharp = \Lambda_-. \tag{60}$$

Their causal difference induces a pairing-compatible quasi-isomorphism from compactly supported covariant data to the smooth global cylinder solution complex. It recovers the fifteen residual endpoint pairs without duplication and transports the $\mathfrak{so}(4,2)$ action to cohomology. Consequently, using Theorem 21.1 as an input,

$$\boxed{H^4_{\mathrm{cov}} \cong H^4_{\mathrm{res}} = \operatorname{span}\{[W_+^2],[W_-^2]\}, \qquad \operatorname{sig} G_{\mathrm{cov}} = (2,0), \qquad G_{\mathrm{cov}} = I_2.} \tag{61}$$

No residual CE matrix is recomputed in this argument.

The causal proof uses a local curvature/Weyl–Cotton presentation to carry the physical helicities, support-local deformation retracts for auxiliary and prolongation variables, and a hybrid algebraic/Green contraction for the propagating endpoint. The positive PDE symmetrizer and the indefinite BV/Krein pairing play different roles: the former gives causal evolution, while the latter is transported through the current and Green identities.

21.4 Relation and combined boundary

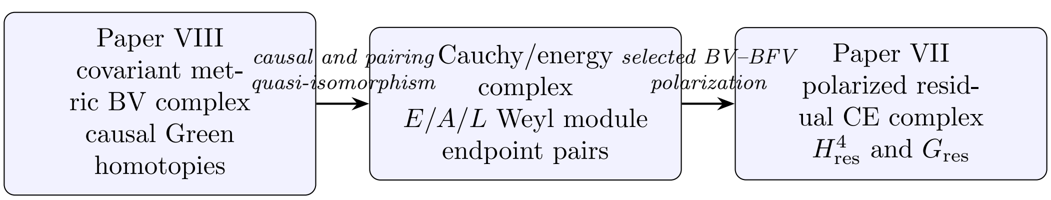

Figure 3: The VII–VIII relation. Paper VII computes the right-hand cohomology. Paper VIII proves that the covariant left-hand complex transports to it; the arrow does not conceal a second CE calculation.

Together the two papers complete the free classical residual branch in its stated category. They do not establish a positive graviton Hilbert space, an integrated $SO(4,2)$ group representation, a preferred alternative boundary polarization, a BRST-compatible Hadamard state, nonlinear stability, a renormalized quantum master equation, anomaly cancellation, or interacting unitarity.

Paper VII determines the selected residual cohomology. Paper VIII proves that the covariant causal metric complex computes and preserves that already determined module and pairing.

22 What was already solved, and what is new

The series does not claim that earlier positive-metric or Krein quantizations failed. Its contribution is to separate their mathematical roles and connect them through an explicit dependency structure.

Paper I [25] reconstructs the Bender–Mannheim positive Pais–Uhlenbeck transformation and classifies the free metric freedom.

Paper II [26] proves the minimum-distortion selection of the canonical positive metric.

Paper III [27] separates metric geometry, vacuum covariance, representation choice, and the Jordan limit in the scalar field.

Paper IV [28] classifies the locally reduced covariant free Einstein–Weyl real forms and their pure-Weyl boundary.

Paper V [29] formulates interaction deformation complexes and identifies resonant scalar obstructions and the surviving doubled boundary structure.

Paper VI [30] applies the deformation test to Einstein–Weyl gravity and isolates the first open gravitational obstruction channel in the stated perturbative class.

Paper VII [31] constructs the selected polarized residual state complex and computes H^4_{res} and its Krein pairing.

Paper VIII [32] constructs causal homotopies for the covariant free metric BV complex and transports Paper VII's theorem, endpoints, symmetry action, and pairing. It does not recompute residual CE cohomology.

> The programme has one local free spine and two branches. Interactions test whether a free completion survives physical conversion. Residual reduction determines the selected state cohomology, and covariant causal transport proves that the metric BV complex computes that same cohomology and pairing.

23 What remains open

Paper VIII closes the free classical covariant causal bridge in the selected closed-universe problem. The construction of causal homotopies, recovery of the residual endpoints, equivariant transport, and comparison of the covariant, Cauchy, energy-mode, and residual pairings are therefore no longer open items. The remaining frontier begins beyond that theorem. The genuine open problems are:

1. distributional and Hadamard completion, including a BRST-compatible microlocal state and renormalized time-ordered products;
2. integration of the infinitesimal $\mathfrak{so}(4,2)$ action to an $SO(4,2)$ representation;
3. alternative boundary, clock, relative, and deparametrized polarizations, in which D or other generators may remain charges;

Paper	Technical role	Claim boundary
I–III	Positive/Krein free scalar geometry	Free oscillator and scalar-field categories
IV	Covariant local free gravity classification	Linearized Einstein–Weyl theory
V–VI	Interaction deformation and obstruction	Stated perturbative orders and channels
VII	Absolute residual H^4 and Krein pairing	Selected algebraic/energy-mode polarization
VIII	Covariant causal BV–BFV transport	Free classical Lorentzian cylinder complex

Table 1: Eight-paper programme grouped by mathematical role.

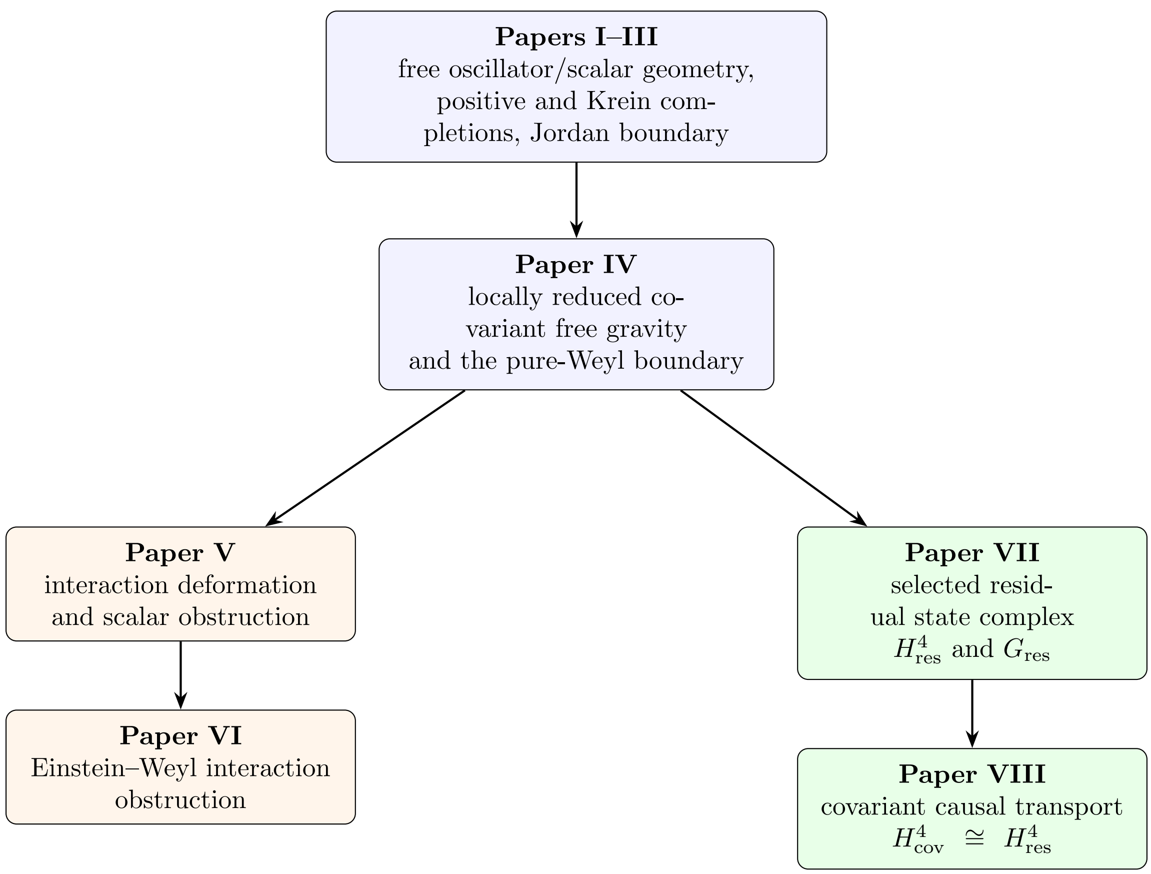

Figure 4: Dependency structure of the eight technical papers. The interaction branch and residual branch share the free gravity spine but answer different questions. Within the residual branch, Paper VII computes the state-side group and Paper VIII proves covariant transport to it.

4. nonlinear BV–BFV transfer and the higher Taub/Kuranishi maps near the conformal cylinder;
5. interaction descent for the two surviving deformation classes;
6. renormalized local composite representatives of W_+^2 and W_-^2;
7. anomaly classification and a renormalized quantum master equation;
8. interacting probability and unitarity.

A Maldacena-type boundary condition may truncate a bulk branch rather than complete the same space [44]; it cannot be modeled by simply deleting one ghost from the selected closed-universe complex. Likewise, classical Green hyperbolicity alone selects neither a Hadamard state nor a quantum product. The quantum problem must in particular control the local anomaly group $H^{1,4}(s \mid d)$, the C^2 beta function, and survival of the residual Cartan homotopy. Proposals for interacting quadratic gravity such as [47] are comparison points, not consequences of Papers VII–VIII.

Do the two classical deformation classes survive an anomaly-free renormalized quantum BV–BFV transfer?

24 Conclusion

The programme separates the higher-derivative ghost question into three levels:

1. Local free classification determines the complex dynamics, real forms, metrics, gauge reduction, and Jordan limits.
2. Gauge and residual reduction determine which state, vertex, or observable classes survive, and covariant causal transport compares their realizations.
3. Interaction stability tests whether a selected free or reduced structure survives physical nonlinear conversion.

Papers I–IV establish the local free spine. The positive PU metric, the minimum-distortion principle, the universal complex covariance, the Krein boundary, and the covariant gravity classification are compatible parts of that spine rather than competing solutions to one undifferentiated problem. Papers V–VI then show that a valid free completion need not admit the same analytic continuation through an interacting branch-changing channel.

Papers VII–VIII complete the selected free classical pure-Weyl branch. Paper VII derives the polarized residual state complex and computes

$$H^4_{\mathrm{res}} = \mathrm{span}\{[W_+^2], [W_-^2]\}, \qquad \operatorname{sig} G_{\mathrm{res}} = (2,0), \qquad G_{\mathrm{res}} = I_2. \tag{62}$$

Paper VIII constructs the support-local curvature prolongation and causal chain homotopies of the covariant metric BV complex, recovers the residual endpoints and symmetry action, and transports the current and Green pairings. It therefore proves

$$H^4_{\mathrm{cov}} \cong H^4_{\mathrm{res}}, \qquad \operatorname{sig} G_{\mathrm{cov}} = (2,0), \qquad G_{\mathrm{cov}} = I_2, \tag{63}$$

using Paper VII's CE theorem as an input rather than recomputing it. The two surviving classes are deformation or vertex classes, not positive-norm one-particle gravitons.

The resulting position is neither that higher-derivative ghosts are automatically harmless nor that they automatically invalidate a theory. The correct question must name the complex dynamics, real form, observable algebra, local and residual quotient, completion, causal realization, and interaction class being considered.

> The eight-paper programme classifies the local free completions, tests their interacting stability, computes the selected pure-Weyl residual cohomology, and proves its covariant causal realization. Its remaining frontier is nonlinear and quantum: alternative boundary problems, Hadamard state selection, the renormalized quantum master equation, anomalies, and unitarity.

Verification. The finite-dimensional, variational, symplectic, causal, and modewise identities quoted here are tied to the verification repositories of Papers I–VIII [25, 26, 27, 28, 29, 30, 31, 32]. Each technical paper states the category and claim boundary of its certificates; a certificate from one level is not evidence for a stronger level.

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